

PAMBAN, India

WMO: 433630

Lat: 9.283N

Lon: 79.217E

Elev: 11

StdP: 101.19

Time Zone: 5.50 (IND)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 24.0	(c) 24.7	(d) 19.3	(e) 14.1	(f) 26.1	(g) 20.1	(h) 14.8	(i) 26.3	(j) 12.1	(k) 26.9	(l) 10.8	(m) 26.1	(n) 1.6	(o) N/A	(p) 0.449

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	5.1	34.7	28.3	34.2	28.1	33.7	28.0	30.0	32.6	29.5	32.3	29.1	32.0	2.1	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.4	26.3	31.8	28.8	25.3	31.4	28.3	24.7	31.2	100.2	32.6	97.6	32.4	95.2	32.1	33.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.9	(b) 9.1	(c) 8.0	(e) 22.1	(f) 36.8	(g) 1.5	(h) 1.2	(i) 21.1	(j) 37.6	(k) 20.2	(l) 38.3	(m) 19.4	(n) 39.0	(o) 18.3	(p) 39.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.5	27.0	27.9	29.6	31.4	31.3	31.1	30.8	30.3	30.0	29.3	28.0	27.1	
	DBStd	1.82	0.75	0.96	1.01	0.91	1.02	0.73	0.72	0.76	0.70	1.32	1.15	0.96	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	7112	526	501	609	643	662	632	646	628	600	597	540	529	
	CDD18.3	4070	268	267	351	393	403	382	387	369	350	339	290	271	
	CDH23.3	50972	2487	2809	4430	5543	5676	5332	5271	4843	4479	4191	3230	2682	
CDH26.7	22860	372	760	1994	3148	3202	2932	2791	2368	2084	1764	954	489		
Wind	WSAvg	2.8	3.4	2.3	1.4	1.7	3.2	3.8	2.9	2.8	2.9	2.2	2.9	3.9	
Precipitation	PrecAvg	920	39	24	22	51	47	2	9	9	33	200	280	193	
	PrecMax	1496	207	308	289	193	195	21	67	50	198	457	523	553	
	PrecMin	357	0	0	0	0	0	0	0	0	0	3	39	12	
	PrecStd	258	52	47	46	52	53	4	17	13	48	114	124	129	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.5	32.4	34.3	35.2	35.2	34.6	34.7	34.2	33.5	33.8	32.2	31.2	
		MCWB	25.3	26.4	27.3	28.5	28.7	27.9	27.6	27.6	27.8	27.7	27.1	26.1	
	2%	DB	29.5	31.3	33.5	34.8	34.5	34.0	33.8	33.2	32.7	32.7	31.2	29.9	
		MCWB	24.9	25.9	27.1	28.4	28.6	28.0	27.6	27.4	27.4	27.4	26.9	25.8	
	5%	DB	28.9	30.5	32.8	34.2	33.9	33.5	33.2	32.7	32.2	32.1	30.5	29.2	
		MCWB	24.7	25.6	27.0	28.3	28.6	28.0	27.6	27.3	27.3	27.2	26.5	25.6	
	10%	DB	28.4	29.8	32.2	33.7	33.3	32.9	32.7	32.1	31.7	31.4	29.8	28.6	
		MCWB	24.5	25.3	26.8	28.1	28.4	28.0	27.5	27.1	27.2	27.0	26.2	25.4	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.7	27.2	29.0	30.2	30.2	30.3	31.0	29.8	29.2	28.8	28.2	27.6
			MCDB	28.8	30.6	32.4	33.5	33.1	32.6	32.8	32.3	32.1	32.0	30.7	29.0
2%		WB	26.1	26.6	28.2	29.5	29.6	29.7	30.1	28.7	28.4	28.2	27.5	26.7	
		MCDB	28.2	30.1	31.8	33.0	32.7	32.2	32.0	31.4	31.5	31.4	30.1	28.7	
5%		WB	25.6	26.2	27.7	29.1	29.2	29.1	29.2	28.1	27.9	27.7	27.0	26.2	
		MCDB	27.8	29.4	31.3	32.7	32.4	31.7	30.7	31.2	31.1	30.9	29.6	28.2	
10%		WB	25.2	25.9	27.3	28.6	28.9	28.6	28.3	27.6	27.6	27.4	26.7	26.0	
		MCDB	27.5	28.9	30.9	32.4	32.1	31.5	30.9	30.8	30.7	30.5	29.1	27.9	
Mean Daily Temperature Range	5% DB	MDBR	3.3	4.4	5.6	5.1	4.3	4.1	4.3	4.3	4.3	4.3	3.5	3.1	
		MCDBR	3.8	5.2	6.2	5.3	4.7	4.6	4.8	4.8	4.6	5.0	4.6	4.0	
		MCWBR	2.0	2.2	2.5	1.9	1.9	2.1	2.2	2.1	1.9	2.0	2.2	2.1	
	5% WB	MCDBR	3.4	4.7	5.5	5.0	4.3	4.3	4.6	4.5	4.5	4.6	4.1	3.5	
		MCWBR	2.0	2.2	2.8	2.2	2.1	2.5	2.9	2.5	2.2	2.2	2.2	2.2	
Clear-Sky Solar Irradiance	taub	0.450	0.447	0.484	0.497	0.563	0.507	0.522	0.506	0.487	0.498	0.493	0.485		
	taud	2.308	2.301	2.203	2.209	2.012	2.204	2.146	2.200	2.262	2.223	2.225	2.221		
	Ebn at Noon	861	876	846	824	755	788	780	805	831	819	814	818		
	Edh at Noon	131	135	151	149	177	144	153	148	140	145	141	140		
All-Sky Solar Radiation	RadAvg	5.20	5.95	6.42	6.27	5.98	5.86	5.83	5.92	5.98	5.21	4.32	4.40		
	RadStd	0.32	0.28	0.29	0.27	0.37	0.29	0.22	0.28	0.30	0.32	0.37	0.34		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.34	-0.16	N/A	+1.01	N/A	-0.30	N/A	N/A	N/A	N/A

Nomenclature: See separate page